# Henry (ChiaChan) Ho

U.S. Permanent Resident (Green Card Holder) – No sponsorship required

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## TECHNICAL SKILLS

Programming Languages: Python, Java, SQL, JavaScript, HTML, MATLAB

Tools: Git, Jupyter, AWS, Vercel, Docker, CI/CD, Pytest, Junit

Libraries/Frameworks: NumPy, Pandas, FastAPI, PyTorch, PostgreSQL, RestfulAPI, RLHF, MongoDB, NoSQL, Scikit-learn, React, Node.js, Flask

## EDUCATION

**University of Pennsylvania** **Expected May 2027**

*Master of Science in Engineering in Data Science and Artificial Intelligence — GPA: 4.0/4.0*

Relevant Coursework: Database & Information Systems, Applied Machine Learning, Graph Neural Network

**University of California, San Diego** **Sep 2022 – Jun 2025**

*Bachelor of Arts in International Business, Minor in Data Science — GPA: 3.85/4.0*

Provost Honors: Fall 2022 – Spring 2024 (5 quarters)

Relevant Coursework: Data Structures & Algorithms, Probabilistic Modeling & Machine Learning, Data Management & Analysis

## EXPERIENCE

**Data Science Intern** **Jun 2026 – Aug 2026**

*(TSMC) Taiwan Semiconductor Manufacturing Company Limited — Hsinchu, Taiwan*

- Built and deployed a full-stack training-analytics platform now used by 23 L&D staff to track NT$150M in annual training budget across 3,500+ courses per year, by independently developing Python data cleaning pipelines, backend services, and an interactive dashboard.
- Restored reliability of the department's pre-class notification system, saving an estimated 911.7 hours (≈0.44 FTE) per year across 13 staff, by fixing four core functions — notification date logic, automated link retrieval, multi-day course templating, and individual-mode delivery.
- Built an automated, data-driven course-scheduling tool that replaced ad-hoc, experience-based scheduling decisions, ensuring adequate course availability for TSMC's 90,000+ employees and supporting execution of the department's NT$150M annual training budget.

**Software Development Engineer** **Dec 2025 – Jun 2026**

*Perelman School of Medicine, University of Pennsylvania — Philadelphia, PA*

- Developed and maintained 10+ RESTful API endpoints for the TissueLab platform, supporting AI-driven whole-slide pathology image analysis workflows and data management.
- Designed a low-code interface builder enabling non-engineers to modularly customize AI model panels via drag-and-drop, and resolved 10+ bugs across backend and frontend systems to improve platform stability and usability.

**AI Research Fellow** **Sep 2025 – Nov 2025**

*Handshake AI Solutions LLC — Remote (Philadelphia, PA)*

- Designed 20+ multimodal reasoning tasks (text + images) applying RLHF, uncovering consistent LLM weaknesses in interpreting hand-drawn diagrams — particularly tree/graph structures and distance-estimation problems.
- Evaluated 60+ model responses across 20+ cases, achieving a 90+% first-pass quality rate, and translated identified failure patterns into structured feedback for model refinement.

**Data Science Instructional Assistant** **Sep 2023 – Aug 2024**

*Halicioglu Data Science Institute — San Diego, CA*

- Taught data science fundamentals to 300+ students and provided 1-on-1 debugging guidance to 100+ students, breaking down core concepts into runnable Python examples and walking through code line-by-line.
- Identified common pain points in student code submissions and collaborated with faculty to redesign assignments, reducing conceptual confusion by over 20%, as reflected in mid-term score improvements.

## PROJECTS

**IdealNest — ZIP-Code-Level U.S. Housing Affordability Web App** **Jan 2026 – May 2026**

*University of Pennsylvania — Philadelphia, PA*

- Built IdealNest, a full-stack web app for exploring U.S. ZIP-code-level housing and neighborhood data, with a FastAPI backend on PostgreSQL (AWS RDS) and a React (Vite, PrimeReact) frontend, deployed on Vercel.
- Implemented a flexible ZIP-search interface covering 10,000+ ZIP codes, with debounced slider- and text-based filtering, four preset "recommended search" queries, and per-ZIP detail drill-down pages.
- Designed the SQL schema and query layer over 200K+ rows of integrated real estate, IRS income, and education data, optimizing query performance by 15%+ through indexing and schema refinements.

**League of Legends Playing Style Analysis** **Mar 2024 – Jun 2024**

*San Diego, CA*

- Engineered a data pipeline in Python using Pandas and NumPy to process 10,000+ professional League of Legends matches, extracting game-level features (e.g., DPM, CKPM) to represent play styles.
- Built and tuned Decision Tree and Random Forest classification pipelines to predict league from in-game aggressiveness features, confirming statistically significant regional playstyle differences ($p < 0.05$) through permutation-based hypothesis testing and fairness evaluation.